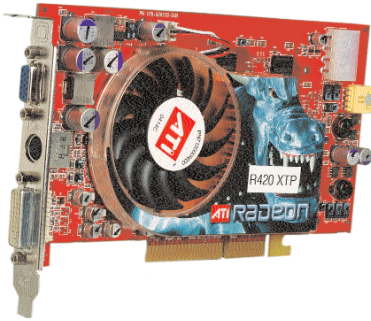




The ATI Radeon X800 XT Platinum features a small heatsink and fan, and both DVI as well as analog outputs

algorithms for anisotropic filtering and full screen anti-aliasing. For video stream processing, the X800 uses the same techniques as do its predecessors, called Videoshader technology. The GPU performs de-interlacing, reduces noise levels, has support for HDTV and applies anti-aliasing, which wipes out the typical artefacts present in MPEG-4 video compression.

nVidia has added a programmable video processor to the NV40, with some features similar to the X800—de-interlacing the video stream, removing blocky artefacts, noise cancellation, and also, support for HDTV. An exciting feature of the programmable video processor is that it can fuse special effects onto the picture, in real-time. Game lighting, for instance, is another exciting feature of the video processor. nVidia has come up with a new technique to render dynamic lighting into computer graphics. The feature should bring in a photorealistic feel into computer graphics, making games more life-like.

The GeForce 6800 Ultra also comes with enhanced algorithms for full screen anti-aliasing, and a new method for anisotropic filtering.

It's the interface too

As of now, both ATi and nVidia are offering AGP versions of the Radeon X800 and the GeForce 6800 Ultra. On the horizon, though, is the new standard in PC I/O—PCI-Express. PCI-Express is expected to debut before the year ends, and will eventually replace the AGP and PCI standards.

It's a matter of bottlenecks—AGP, for instance, was developed to work around the inadequacies of PCI. But if you take applications such as HDTV-video editing, you've just about crossed into a zone that needs the kind of I/O bandwidth that PCI-Express offers. Over the coming months, both nVidia and ATi will offer solutions that will make the transition to PCI-Express from the AGP 8X interface.

What does that mean for graphics cards? They now won't be limited by slower speeds due to the interconnect between the system I/O and the graphics subsystem. It also means that cards can now happily consume more power to run—the GeForce 6800 FX specs already necessitates a 480 W power supply. Industry analysts say that with PCI-Express coming into a position to overtake AGP, card makers will switch over to it. ATi and nVidia are working on bridge interconnects that allow for a smoother transition.

So what's in store?

Processing geometric data for today's games needs fascinating, cutting-edge hardware; nVidia and ATi are developing just such fancy hardware. As CPUs get ever more powerful, system memory moves to DDR2, and PCI-Express makes a move into your desktop, these advanced graphics processing beasts are going to push the limits of gaming with fluid, fast gameplay, even when all the bells and whistles are turned on. It's still going to be a game, and not reality—not yet, at least—but you'll come as close as you ever have with the latest offerings. Gentlemen, start your engines. Let the games begin. 

SRINIVASAN RAMAKRISHNAN

srinivasan_ramakrishnan@thinkdigit.com

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